Introduction to Limits

Evaluating limits

1 Evaluate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression.

2 Evaluate by direct substitution where possible:

a $\lim_{x \to 3} (x^2 + 2x - 1)$

b $\lim_{x \to 1} \frac{x+5}{x+1}$

3 Using a table of values near $x = 0$ (in radians), estimate $\lim_{x \to 0} \frac{\sin x}{x}$.

One-sided limits and limits at infinity

4 Let $f(x) = \frac{|x|}{x}$. Find:

a $\lim_{x \to 0^+} f(x)$

b $\lim_{x \to 0^-} f(x)$

c $\lim_{x \to 0} f(x)$

5 Evaluate each limit at infinity:

a $\lim_{x \to \infty} \frac{3x^2 + 5}{x^2 - 2}$

b $\lim_{x \to \infty} \frac{2x+1}{x^2 + 4}$

6 The graph of $f(x) = \frac{1}{x - 2} + 3$ has asymptotes $x = 2$ and $y = 3$. Express each asymptote as a limit statement.

More limit computations

7 Evaluate $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$ by rationalizing the numerator.

8 Evaluate $\lim_{x \to -1} \frac{x^3 + 1}{x + 1}$ by factoring (using the sum-of-cubes formula).

9 Evaluate $\lim_{x \to 0} \frac{\tan x}{x}$, using $\lim_{x \to 0} \frac{\sin x}{x} = 1$ and $\tan x = \frac{\sin x}{\cos x}$.

Continuity and limits together

10 A function f is defined by $f(x) = x + 3$ for $x \neq 2$, and $f(2) = 10$. Find $\lim_{x \to 2} f(x)$ and state whether f is continuous at $x = 2$.

11 Estimate $\lim_{x \to \infty} \left(\sqrt{x^2 + 1} - x\right)$ by rationalizing.

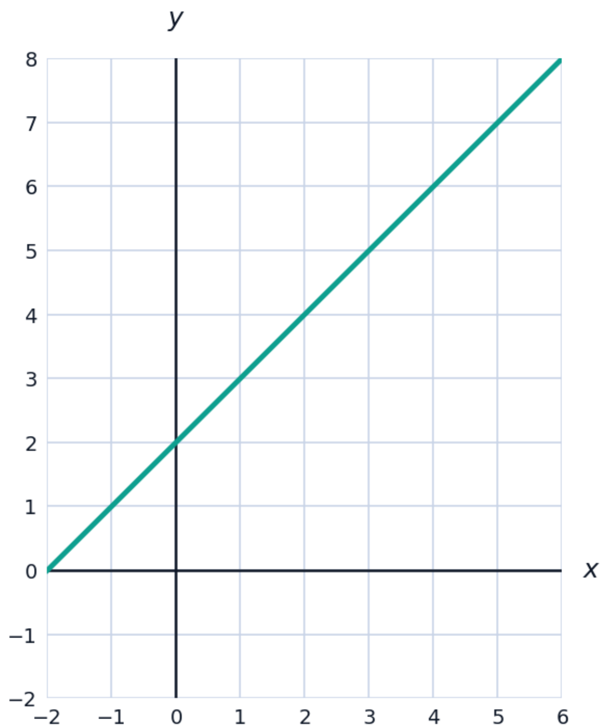
Limits from graphs and tables

12 A function g satisfies $g(1.9) = 5.8$, $g(1.99) = 5.98$, $g(1.999) = 5.998$, $g(2.1) = 6.2$, $g(2.01) = 6.02$, $g(2.001) = 6.002$. Based on this table, estimate $\lim_{x \to 2} g(x)$.

13 Explain, using one-sided limits, why $\lim_{x \to 0} \frac{1}{x^2}$ exists as an infinite limit (both one-sided limits agree), while $\lim_{x \to 0} \frac{1}{x}$ does not exist even as an infinite limit.

Visualizing a limit graphically

14 The graph of $f(x) = \frac{x^2 - 4}{x - 2}$ (which simplifies to $x + 2$ with a hole at $x = 2$) is shown. Use the graph to estimate $\lim_{x \to 2} f(x)$, and explain why the hole doesn't affect the limit's value.



The formal idea of a limit (intuitive)

15 Explain, in your own words, what it means for $\lim_{x \to a} f(x) = L$, without using formal epsilon-delta language.

Limit laws

16 Given $\lim_{x \to 3} f(x) = 5$ and $\lim_{x \to 3} g(x) = -2$, use the limit laws to evaluate:

a $\lim_{x \to 3} [f(x) + g(x)]$ **b** $\lim_{x \to 3} [f(x) \cdot g(x)]$ **c** $\lim_{x \to 3} \frac{f(x)}{g(x)}$

Limits involving piecewise functions

17 A function is defined by $f(x) = x^2$ for $x < 1$ and $f(x) = 3x - 1$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.